

2a	The resultant force acting on a body is the rate of change of momentum of the body and acts in the direction of the change in momentum.	E	A1
2b	Work W is the product of a force F and a displacement s in the direction of the force. Since Power P is the rate of doing work, Hence, $P = W/t$ $= Fs/t$ $= Fv$	A	B1 B1 A0

2ci	At constant speed, <u>no net force acting on lorry.</u> Hence, magnitude of <u>driving force is equal to that of resistive force</u> on lorry. Driving force, $F = P/v$ $= 130\,000/25$ $= 5200\text{ N}$	A	M1 M1 A0
2cii	<u>Friction</u> between tyres of lorry and road exerts a <u>backward force on the road.</u> By Newton's third law, road surface exert a <u>forward force</u> of <u>equal magnitude on tyres.</u>	A	B1 B1
2d	$F - mg \sin \theta - R = ma$ $F = 36000(9.81 \sin 1.4) + 5200 + (36000)(0.15)$ $= 19228\text{ N} = 19000\text{ (19200) N}$	A	C1 A1